

## Laws of exponents



1 Simplify the following expressions:

a  $a^s \times a^t$

b  $\frac{a^s}{a^t}$

c  $a^s \times a^s$

d  $a^s + a^s$

e  $a^s + a^t$

f  $2^2 \times 2^2$

g  $4y^3 \times 6y$

h  $3x^5 \times 8x^2$

2 Simplify the following expressions:

a  $\frac{x^6}{4x^4}$

b  $p^{18} \div p^8 \div p^5$

c  $\frac{4n^3 \times 4n^4}{16}$

d  $\frac{u^{2x+1}}{u^x}$

e  $p^7 \div p^3 \times p^5$

f  $p^{14} \div p^6 \div p^5$

g  $\frac{y^7 \times y^5}{y^3 \times y^2}$

h  $12s^8 \div 3t^6 \times 6t^4$

i  $\frac{10p^6 \times 3p^{10}}{15p^2}$

j  $c^{2z} c^{z+1}$

k  $20m^6 \div 5m^{13} \times 9m^2$

l  $\frac{12y^4 \times 35y^8}{4y^5 \times 5y^4}$

3 Complete the statement below:

$$15j^{14} \div (\quad) = 5j^7$$

4 Rewrite the following expressions without parantheses:

a  $\left(\frac{a}{b}\right)^3$

b  $(xy)^7$

c  $(2x)^4$

d  $\left(\frac{xy}{5z}\right)^2$

5 Write  $(16^p)^4$  in the form  $a^b$ , where  $a$  is a prime number.

6 For each of the following expressions, state the operation that should be done to the exponents to simplify the expressions:

a  $(z^{28})^{19}$

b  $z^{13} \times z^{16}$

c  $z^{14} \div z^{13}$

d  $\frac{z^{26}}{z^{13}}$

7 Simplify the following expressions:



a  $\frac{(x^3)^2}{x^3}$

c  $p^2 \times (p^3)^2$

e  $\frac{(x^2)^6}{(x^2)^2}$

g  $\frac{(4y^3)^5}{(y^2)^2}$

i  $(u^9 \times u^5 \div u^{19})^2$

k  $\frac{(-35m^4) \times (3m^4)^3}{7m^3 \times (m^2)^5}$

m  $(t^6)^5 \div \left(\frac{t^9}{t^7}\right)^3$

b  $(u^{x+1})^3$

d  $p^8 \div (p^2)^2$

f  $(2y^4)^2 \times (2y^2)^3$

h  $(3y^2)^3 \times (-2y^2)^2$

j  $\frac{28m^4 \times (3m^3)^3}{7m^2 \times (m^2)^2}$

l  $\left(\frac{t^4}{t^2}\right)^5 \div \frac{t^5}{t^8}$

n  $\frac{(u^{x+3})^3}{u^{x+1}}$

## Zero and negative exponents

8 Simplify:

a  $a^0$

c  $(2 \times 13)^0$

e  $(13x^7)^0 + 13^0 - 13h^0$

b  $8x^0$

d  $(a^0)^{79}$

f  $(7m^0 + 4)^2$

9 Express the fraction  $\frac{m}{n^4}$  as a product using negative exponents.

10 Express the following in simplest form without negative exponents:

a  $4a^{-2}$

e  $m^{-5}n^{-4}p^4$

b  $(3a)^{-1}$

f  $2y^9 \times 3y^{-5}$

c  $6x^{-3}y^4$

g  $\frac{9x^3}{3x^{-4}}$

d  $m^2 \times m^{-7}$

h  $\frac{12x^3}{4x^7}$



**11** Express the following in simplest form without negative exponents:

**a**  $\left(\frac{a}{b}\right)^{-5}$

**b**  $(4m^{-10})^4$

**c**  $(4m^{-8})^{-3}$

**d**  $(5y^3)^{-3}$

**e**  $\left(\frac{x^2}{y^4}\right)^{-1}$

**f**  $\left(\frac{a^3}{b^3}\right)^{-5}$

**g**  $\left(\frac{2h}{3}\right)^{-4}$

**h**  $\frac{(m^{-3})^{-1} \times (m^4)^{-3}}{m^3 \times m^4}$

**12** Express the following in simplest form without negative exponents:

**a**  $2p^4q^{-2} \times 5p^{-4}q^{-5}$

**b**  $\frac{5p^5q^{-4}}{40p^5q^6}$

**c**  $10x^3y^2z \div 2x^5y^9$

**d**  $2^{15} \div 2^{-5}$